

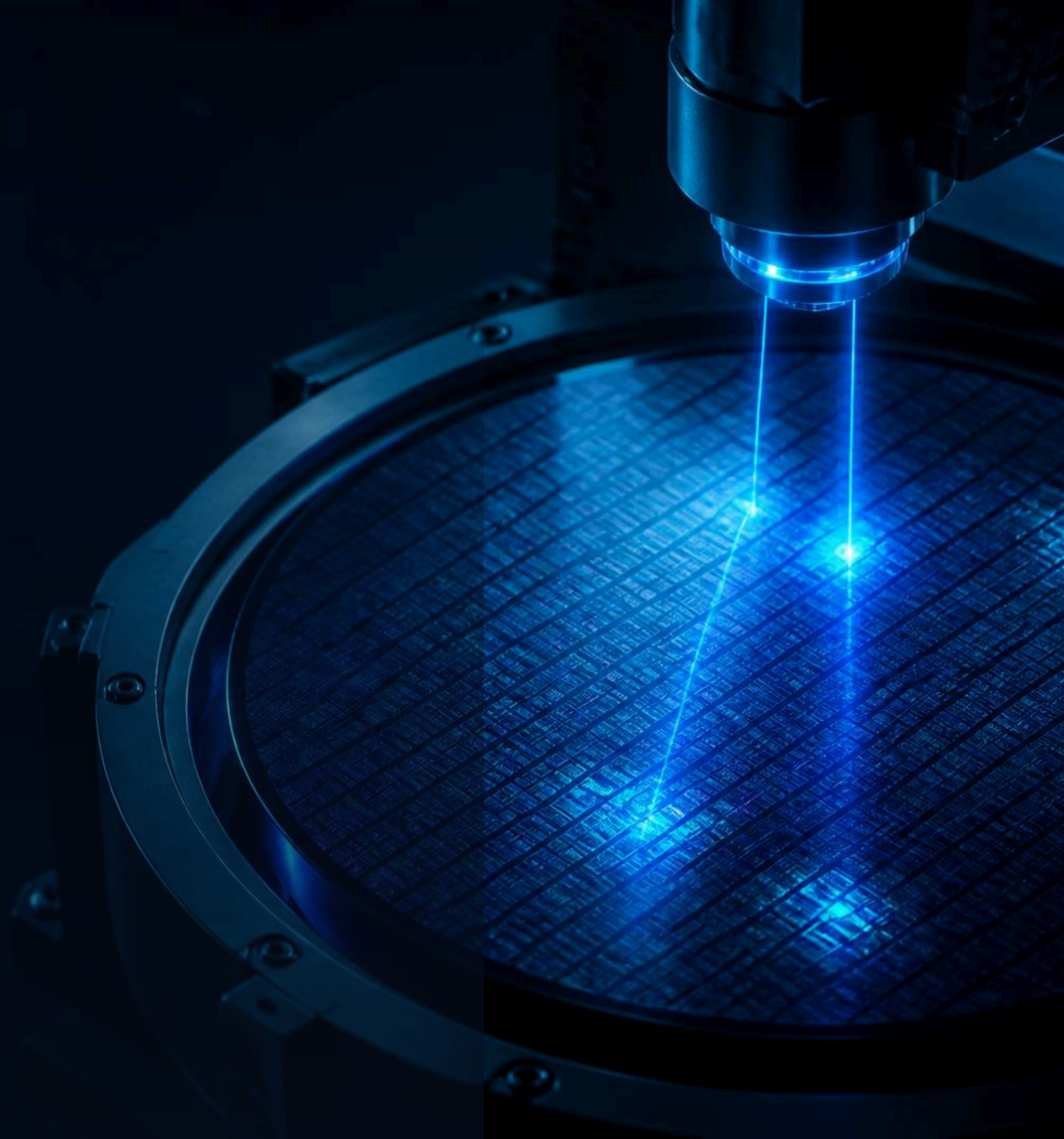
VISIONX × IBM BOB AI HACKATHON

S1 ANALYZER

**Wafer Yield Root Cause &
Defect Pattern Analyser**

AI-powered defect detection, process-risk prediction,
root-cause analysis and preventive batch screening.

TEAM / VISIONX



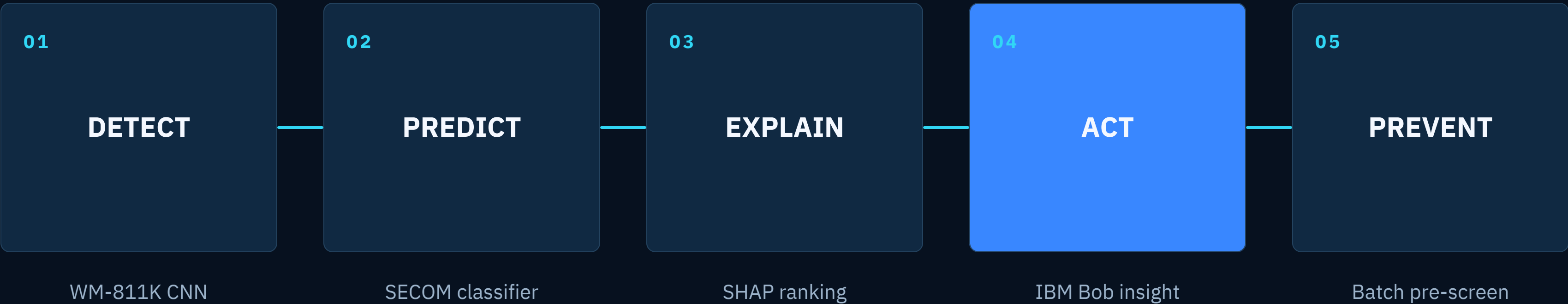
Yield risk hides across disconnected signals.

At advanced nodes, small process deviations can create expensive, hard-to-diagnose losses.



One evidence chain. Five engineering moves.

A multimodal pipeline converts wafer imagery and process data into preventive decisions.



Traceable from input to engineering output.

INPUT DATA

Wafer images

Process parameters

Historical records

Upcoming batches

API LAYER

FastAPI service

ML + XAI

PyTorch CNN

XGBoost VotingClassifier

SHAP analysis

INTELLIGENCE

Risk engine

IBM Bob / watsonx.ai

OUTPUT

Defect pattern

Process risk

Root causes + actions

Upcoming risk

Python • FastAPI • PyTorch • scikit-learn • XGBoost • SHAP • watsonx.ai • PostgreSQL • SQLAlchemy • Alembic

Two trained models. Inference-only by design.

The current implementation loads established artifacts; it does not retrain either model.

MODEL 01 / WAFER

one_vs_one_classifier_model.pth

PyTorch CNN classifier
Input / wafer image or mapped record
Output / defect pattern class
Dataset / WM-811K

Evaluation metric: not available in current implementation

MODEL 02 / PROCESS

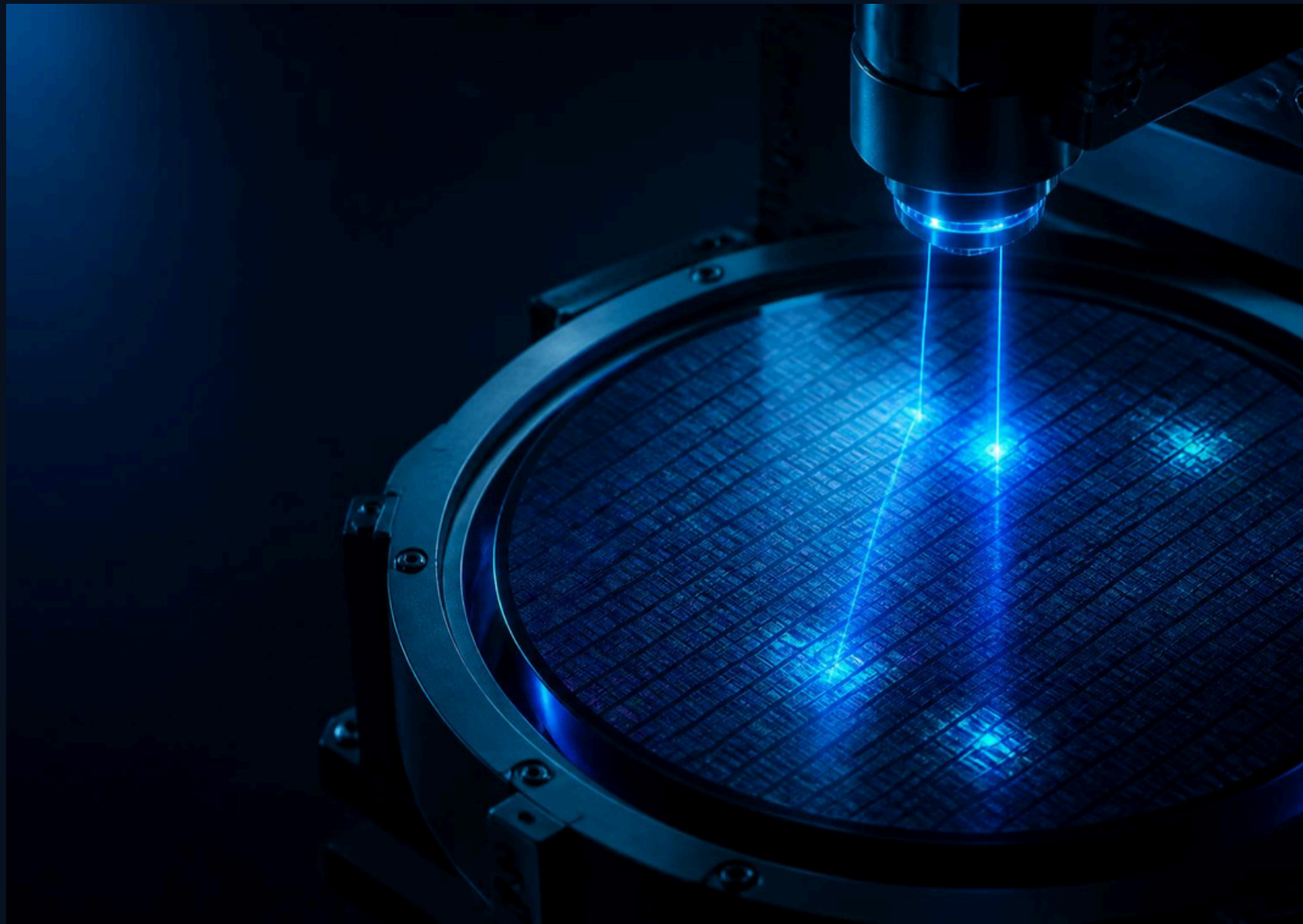
secom_classifier_model.pkl

XGBoost VotingClassifier
Input / process sensor parameters
Output / risk class + probability when supported
Dataset / SECOM

Evaluation metric: not available in current implementation

Classify the wafer defect pattern.

A trained PyTorch CNN processes a wafer image or mapped wafer record and returns a pattern class.



EXAMPLE RUNTIME ANALYSIS

Evidence, not invented certainty.

The application presents the predicted class. Confidence probabilities appear only when the loaded model supports them. No hard-coded prediction or confidence value is used here.

APPLICATION CAPTURE AREA
Connect verified runtime screenshot

Surface process risk before inspection.

SECOM process parameters are evaluated by the XGBoost VotingClassifier.



* Probability is displayed only when supported by the model artifact. No fabricated metrics or values.

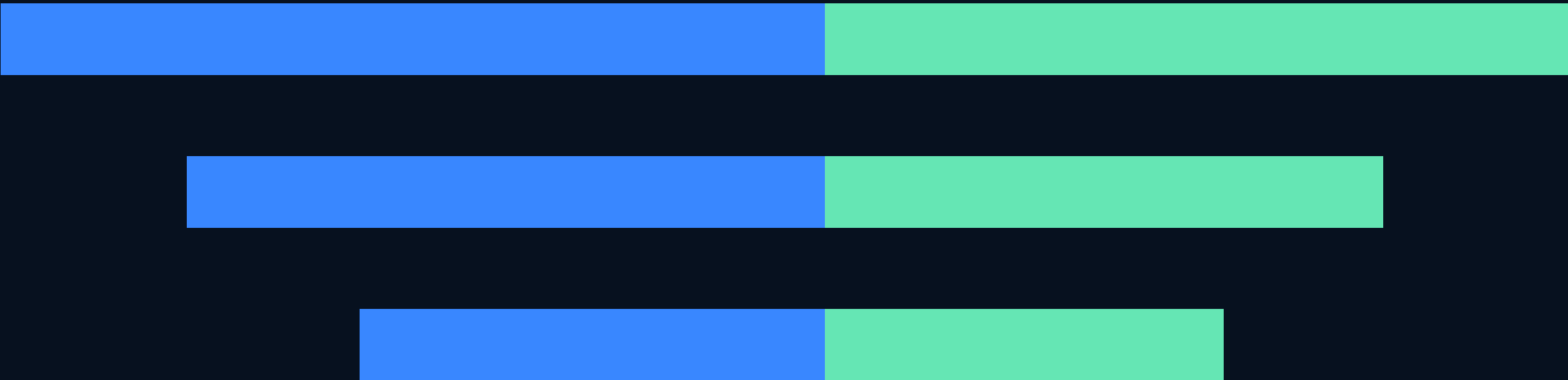
ENGINEERING BENEFIT / Earlier visibility into potential process failure—before physical wafer inspection.

Rank what pushed risk up—or down.

SHAP converts the process model output into feature-level contribution evidence.

↑ INCREASES RISK

↓ REDUCES RISK



SECOM features may be anonymized. Physical sensor meaning requires manufacturing metadata that is not included; no sensor interpretation or SHAP value is fabricated.

Translate model evidence into engineering action.

IBM Bob receives predictions, SHAP explanations and risk context—not raw authority.

EVIDENCE IN

- ML predictions
- SHAP contribution ranking
- Risk context

IBM Bob does not replace the ML models or assert unsupported causality.

IBM BOB / EVIDENCE-GROUNDED

Assessment
Evidence summary
Contributing factors
Corrective actions
Next checks

Screen the next batch before execution.

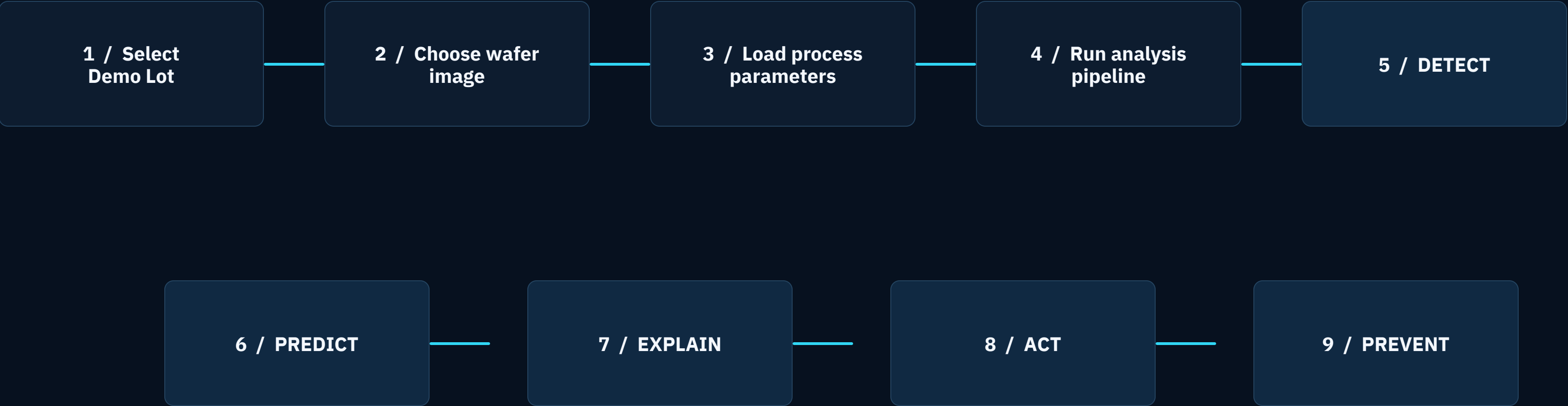
Scheduled process parameters are scored with the same SECOM risk model.

BATCH ID	SCHEDULED	RISK STATUS	RISK SCORE	TOP FACTORS
Demo batch mapping		Runtime output only		No live fab claim
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Application/demo-level batch mappings—not live manufacturing integration.

From lot selection to decision.

Supported entry modes can include Demo Lot, Wafer Image, Process CSV and New Lot.



Built for reproducible AI analysis.

Repository scripts, configuration and migrations support consistent environment setup.



PYTHON / FASTAPI / PYTORCH / SCIKIT-LEARN / XGBOOST / SHAP / WATSONX.AI / POSTGRESQL

Clear service boundaries keep inference, explanation, interpretation and persistence independently testable.

Explainable yield intelligence—without overstating the evidence.



VISIONX S1 ANALYZER

Turning semiconductor process data into explainable, actionable yield intelligence.

CURRENT LIMITS / Independent WM-811K + SECOM datasets • application-level demo mapping • anonymized features • inference-only • no live fab control

NEXT / Live fab data • richer sensor mapping • GPU scaling • deeper IBM Bob integration

THANK YOU / Q&A